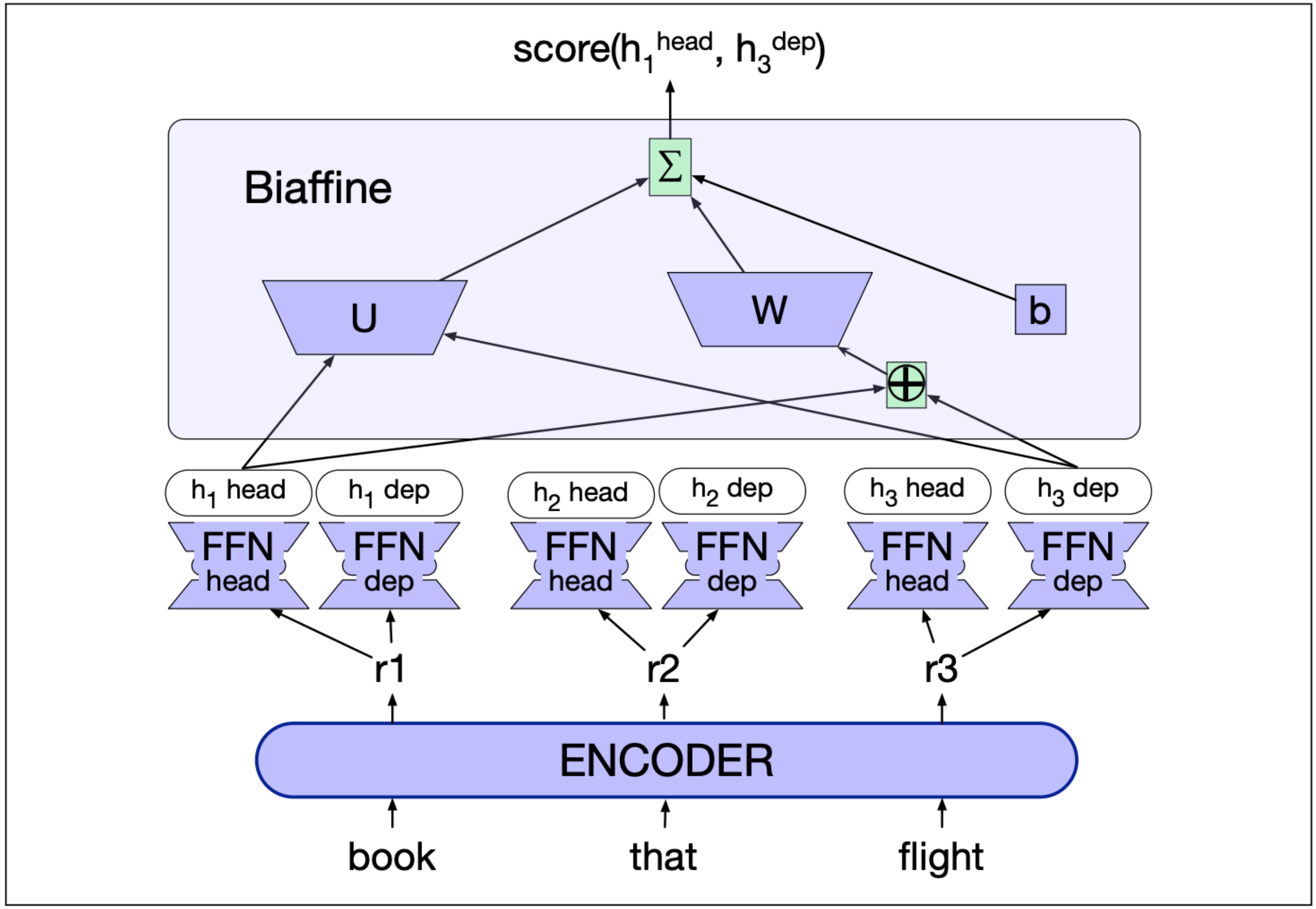
Neural MST Parsing

- Dozat and Manning (2017) proposed using MST only during inference, and to train edge scores using neural networks
- The words are first encoded using a bi-RNN encoder, which produces a word embedding for each word $r_1,\ldots,r_n$
- Each embedding is fed into two networks, producing two series $h_i^{(head)}$ and $h_i^{(dep)}$



Deep Biaffine Attention for Neural Dependency Parsing. Dozat and Manning, ICLR 2017

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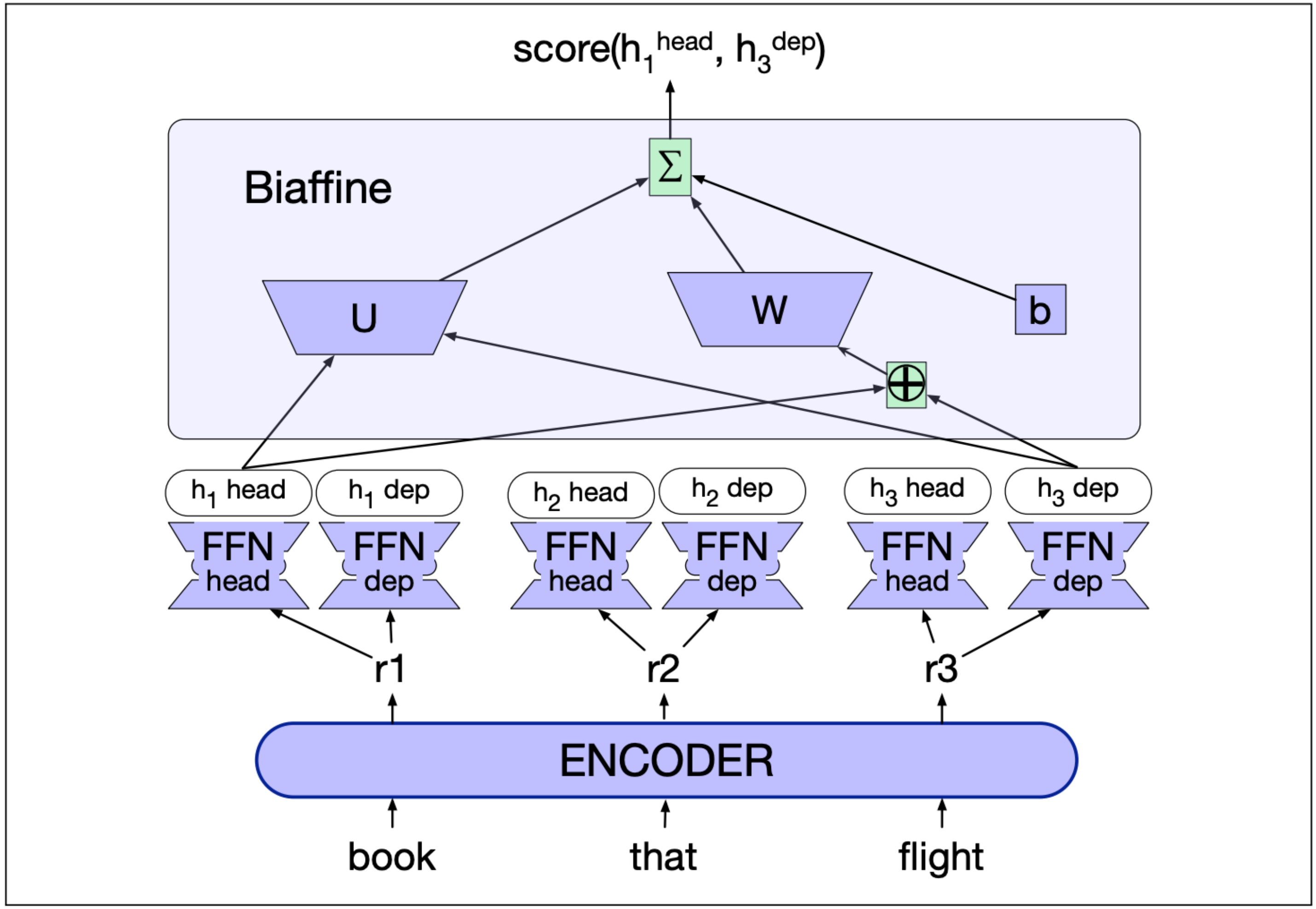

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Neural MST Parsing: Unlabeled Variant

- We then define a distribution for each word *j*, a distribution over its potential headwords *i*

$$
\begin{aligned}
\mathrm{Score}(i \rightarrow j) &= \mathrm{Biaff}(\mathbf{h}_i^{head}, \mathbf{h}_j^{dep}) \\
\mathrm{Biaff}(\mathbf{x}, \mathbf{y}) &= \mathbf{x}^\mathsf{T}\mathbf{U}\mathbf{y} + \mathbf{W}(\mathbf{x} \oplus \mathbf{y}) + b
\end{aligned}
$$

$$p(i \rightarrow j) = \mathrm{softmax}([\mathrm{Score}(k \rightarrow j); \forall k \neq j, 1 \leq k \leq n])$$

- Training is done locally (i.e., the prediction of a word’s head is viewed as independent from the prediction of other words’ heads.

Neural MST Parsing: Labeled Variant

- Inference is done through standard MST inference

- For labeled parsing, Dozat and Manning train two classifiers: the one seen last slide, and another label-scorer, which has the same structure, but whose prediction is:
  - Given a dependency edge, predict a softmax over the possible dependency label
  - Inference: first predict the unlabeled dependency, then given the predicted edge, predicts its most likely label

Some Results

- The basic MST parser scores 90.9% LAS on English (Wall Street Journal articles, in domain setting)
- Koo & Collins 2010, third-order dependency parser: ~93% UAS on the same domain
- Dozat & Manning 2017, deep biaffine parser: 95.7 UAS / 94.1 LAS.

- Small, albeit consistent gains, from using higher-order features with neural representation

https://aclanthology.org/2020.acl-main.776.pdf